Chapter 1 • Real Numbers

Solved Review Exercise 1

MCQs

1. √7 is:
a) integer
b) rational number
c) irrational number
d) natural number
Ans: c
7 is not a perfect square.

2. π and e are:
a) natural numbers
b) integers
c) rational numbers
d) irrational numbers
Ans: d
Both are non-terminating, non-recurring constants.

3. If n is not a perfect square, then √n is:
a) rational number
b) natural number
c) integer
d) irrational number
Ans: d
For positive integer n, a non-perfect square has an irrational square root.

4. √3 + √5 is:
a) whole number
b) integer
c) rational number
d) irrational number
Ans: d
The sum does not simplify to a rational number.

5. For all $x \in \mathbb{R}$, x = x is called:
a) reflexive property
b) transitive property
c) symmetric property
d) trichotomy property
Ans: a
Every real number equals itself.

6. If a > b and b > c, then a > c by the:
a) trichotomy property
b) transitive property
c) additive property
d) multiplicative property
Ans: b
Order passes through the intermediate quantity b.

7. If $2^x \times 8^x = 64$, then x =
a) $\frac{3}{2}$
b) $\frac{3}{4}$
c) $\frac{5}{6}$
d) $\frac{2}{3}$
Ans: a
$2^x \cdot 2^{3x} = 2^6$, so $4x = 6$.

8. The statement a = b and b = a illustrates the:
a) reflexive property
b) symmetric property
c) transitive property
d) additive property
Ans: b
Equality may be reversed.

9. √75 + √27 =

studysapiens.com

a) √102
b) 9√3
c) 5√3
d) 8√3
Ans: d
√75 = 5√3 and √27 = 3√3.

10. The product (3 + √5)(3 − √5) is:
a) prime number
b) odd number
c) irrational number
d) rational number
Ans: d
The product is 9 − 5 = 4.

2. For a = $\frac{3}{2}$, b = $\frac{5}{3}$ and c = $\frac{7}{5}$, verify both distributive forms.
(i) a(b+c) = (3/2)(46/15)
= $\frac{23}{5}$
ab+ac = $\frac{5}{2}$ + $\frac{21}{10}$
= $\frac{23}{5}$
(ii) (a+b)c = (19/6)(7/5)
= $\frac{133}{30}$
ac+bc = $\frac{21}{10}$ + $\frac{7}{3}$
= $\frac{133}{30}$
LHS = RHS in both cases.

3. For a = $\frac{4}{3}$, b = $\frac{5}{2}$ and c = $\frac{7}{4}$, verify associativity for addition and multiplication.
Addition: a+(b+c) = $\frac{4}{3}$ + $\frac{17}{4}$
= $\frac{67}{12}$
(a+b)+c = $\frac{23}{6}$ + $\frac{7}{4}$
= $\frac{67}{12}$
Multiplication: a(bc) = (4/3)(35/8)
= $\frac{35}{6}$
(ab)c = (10/3)(7/4)
= $\frac{35}{6}$
Both associative properties verified.

4. Is 0 a rational number? Explain.
Yes. For example, 0 = $\frac{0}{1}$, where 0 and 1 are integers and the denominator is non-zero.

5. State the trichotomy property.
For any real numbers a and b, exactly one of the relations a < b, a = b or a > b is true.

6. Find two rational numbers between 4 and 5.
First average = $\frac{4+5}{2}$ = $\frac{9}{2}$ Second average = $\frac{4+\frac{9}{2}}{2}$ = $\frac{17}{4}$
$\frac{9}{2}$ and $\frac{17}{4}$ (other valid answers are possible).

7. Simplify the three radical/index expressions.
(i) $(x^{15}y^{35}/z^{20})^{\frac{1}{5}} = x^3y^7z^{-4}$.
x^3y^7/z^4
(ii) $\sqrt[3]{(27^{2x})} = (3^{6x})^{\frac{1}{3}}$
$= 3^{2x}$.
(iii) $6\cdot3^{n+2}/(3^{n+1}-3^n) = 54\cdot3^n/(2\cdot3^n)$

studysapiens.com

= 27.

8. The sum of three consecutive odd integers is 51. Find them.

Let the integers be n, n+2, n+4.
3n + 6 = 51
⇒ n = 15.
15, 17 and 19.

9. Ninety-six balls are placed in two buckets; one has 28 more than the other. Find both counts.

Let the smaller count be x; then x + (x+28) = 96.
2x = 68 ⇒ x = 34; the other bucket has 34 + 28 = 62.
34 balls and 62 balls.

10. Salma invested Rs. 350,000 at 7¼% simple profit for 2 years, then 8% for the next 5 years. Find the final amount.

First profit = 350,000 × 2 × 0.0725 = Rs. 50,750.
Second profit = 350,000 × 5 × 0.08 = Rs. 140,000.
Amount = 350,000 + 50,750 + 140,000.
Rs. 540,750.

studysapiens.com